

The classifier outputs:

Digit 0: 1

Digit 1: 2

Digit 2: 8

Digits 3 through 9: 0

Step 1: Exponentiate every logit

$$e^1 \approx 2.72$$

$$e^2 \approx 7.39$$

$$e^8 \approx 2980.96$$

$$e^0 = 1$$

Since there are seven remaining classes with logit 0:

$$7 \times 1 = 7$$

So we now have:

Digit 0: 2.72

Digit 1: 7.39

Digit 2: 2980.96

All others together: 7

Step 2: Add them together

$$2.72 + 7.39 + 2980.96 + 7$$

$$\approx 2998.07$$

This is the “sum-exp” part.

Step 3: Take the logarithm

$$\log(2998.07) \approx 8.006$$

Notice that this is very close to the largest logit, which was 8.

That is why log-sum-exp is often described as a smooth version of the maximum.

Step 4: Add the negative sign

$$\text{Energy} \approx -8.006$$

So this input receives low energy because the classifier has one very strong logit.

The interpretation is:

The network strongly believes this looks like digit 2



Large logit for digit 2



Large log-sum-exp



Negative of that value is low energy

What does softmax produce?

The probability for digit 2 is approximately:

$$2980.96 / 2998.07 \approx 0.9943$$

So:

Digit 2 probability $\approx 99.43\%$

The others are approximately:

Digit 0: 0.09%

Digit 1: 0.25%

Each remaining digit: 0.03%

So the same logits provide two views:

Softmax:

Which digit is most likely?

Answer: digit 2

Energy:

How strong is the total class evidence for this image?

Answer: very strong, so the energy is low

The energy is not based only on the winning class, but because the logit 8 dominates the others, it ends up very close to negative 8.